

Answer the following questions from the Textbook. 10 points for each question

1 (10 points). Are there dependencies between any of the security concepts that we covered? For example, is authentication required for authorization? Why or why not?

2 (10 points). Why is two-factor authentication better than one-factor authentication?

3 (10 points). A Trojan horse is a program that advertises that it does one thing, but really does something that's malicious. For example, a computer chess program that deletes the contents of the user's hard drive when the user loses the game is a Trojan horse. How would the Bell-LaPadula model prevent an attacker that only has unclassified access to a system from obtaining top secret information even if the attacker is able to trick his boss (who has top secret clearance) to run a Trojan horse?

4 (10 points). State the difference between confidentiality and data integrity in three sentences or less.

5 (10 points). What happens if a client connects to SimpleWebServer, but never sends any data and never disconnects? What type of an attack would such a client be able to conduct?

6 (10 points). HTTP supports a mechanism that allows users to upload files in addition to retrieving them through a PUT command.

a (5 points). What threats would you need to consider if SimpleWebServer also had functionality that could be used to upload files?

b (5 points). For each of the specific threats you just listed, what types of security mechanisms might you put in place to mitigate the threats?

7 (10 points). Rewrite the `serveFile()` method such that it imposes a maximum file size limit. If a user attempts to download a file that is larger than the maximum allowed size, write a log entry to a file called `error_log` and return a "403 Forbidden" HTTP response code (**5 points**). What happens if an attacker tries to download `/dev/random` after you have made your modification (**5 points**)?